

A Lean-Verified Fresh-Sample Stochastic Extension of Negative-Weighted Distributionally Robust Optimization

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Accompanying mathematical note for the Lean formalization

Abstract

This note develops a finite-dimensional online stochastic extension of Negative-Weighted Distributionally Robust Optimization (NegDRO). The method combines nearest-point-projected stochastic descent in the prediction parameter with exponentiated-gradient ascent in the environmental weights; both updates may use the same fresh observation. Under a corrected realized additive-intervention reduced-form model, a universal fresh conditional law, fourth-moment bounds, a bounded closed convex feasible set, and a fixed spectral heterogeneity witness, we prove explicit finite-time and averaged-iterate expected squared-error bounds. The complete conditional proof chain is verified in Lean 4 and Mathlib, including the realized model bridge, oracle unbiasedness and integrability, Euclidean projection, entropy regret, spectral curvature, and telescoping recursion. With $\eta_b = \eta_w = T^{-1/2}$, the optimization contribution is of order $T^{-1/2}$ in expected squared error; the penalized result has the additional structural bias $2\mu/\lambda$. The analysis is based on a fixed comparator and does not require independence between the two stochastic gradients. It is an extension developed in this note, not a literal formalization of the empirical-risk exact-maximization Algorithm 1 in arXiv:2412.11850v3 [1].

1 Introduction and Background

1.1 NegDRO and causal invariance

Multi-environment data can distinguish a stable causal relationship from environment-specific predictive associations. Wang, Hu, Bühlmann, and Guo [1] formulate this idea as Negative-Weighted Distributionally Robust Optimization (NegDRO), a continuous minimax problem whose effective environmental weights may be negative. The negative weights penalize disparities between environmental risks and thereby encourage risk invariance. Under additive interventions and sufficient environmental heterogeneity, their population analysis relates stationarity of the NegDRO objective to recovery of the causal outcome coefficient β^* .

The official algorithms in arXiv:2412.11850v3 optimize empirical risks from a fixed data set. In particular, official Algorithm 1 computes an exact penalized maximizer in the environmental weight and then takes a primal gradient step. The present work addresses a different question: can one use one fresh environmental observation per round, update the primal and dual variables simultaneously, and obtain a direct population error bound? The answer given here is conditional on an explicit fresh-sampling law and moment, geometry, and heterogeneity assumptions.

1.2 The stochastic extension

At each round, our method evaluates both stochastic oracles at the current predictable state (b_t, w_t) . It takes a projected descent step in b and an exponentiated-gradient ascent step in w . The two oracles may be formed from the same observation; their mutual independence is neither assumed nor used. The analysis compares w_t with a fixed simplex vector w^0 whose heterogeneity matrix has eigenvalues bounded below by $\lambda > 0$. This witness is an analytical comparator, not the algorithmic initialization.

The proof has a common structure in the penalized and unpenalized cases:

$$\begin{aligned} \text{corrected risk expansion} &\implies \text{fixed-witness curvature} \implies \text{comparator identity} \\ &\implies \text{entropy regret} \implies \text{projected primal recursion.} \end{aligned}$$

The unpenalized comparator identity transfers the dual regret with a factor of two. The penalized identity adds exactly the direction loss 2μ , which becomes the squared-error bias $2\mu/\lambda$ after the curvature division. Instantaneous comparator regret is not assumed to be nonnegative.

1.3 Contributions and verified scope

This note makes the following contributions.

- It states a corrected realized reduced form for additive interventions and derives the common population-risk expansion used by the online analysis. The sign convention is forced by direct elimination from the structural equations; the discrepancy with the printed official Appendix E.1 identity is documented in Section 2.2.
- It analyzes a fresh-sample stochastic projected-SGD/EG recursion on a general nonempty closed convex bounded set. The universal conditional sampling law yields the required selected moments, conditional oracle means, and oracle integrability.
- It gives exact finite-time bounds for the time-average of expected squared errors and for the expected squared error of the averaged iterate. Equal steps $\eta_b = \eta_w = T^{-1/2}$ yield explicit inverse-square-root optimization terms.
- It provides a complete Lean-verified conditional proof of this chain. Completeness is relative to the stated model, sampling, moment, spectral, and feasible-set hypotheses; it is not an assumption-free claim or an external certification.

The result is neither a last-iterate nor a high-probability theorem. A separate square-root consequence concerns root mean square error and must not be read as a bound on expected norm. The project also does not construct an infinite independent sample stream or a product probability space. These scope boundaries are summarized in Section 5.

We use arXiv:2412.11850v3, revised January 7, 2026, as the source version of the original paper. Statements attributed to that paper and statements proved for the stochastic extension are distinguished throughout.

2 Model, Oracle, and Algorithm

We write $\mathcal{E} = \{1, \dots, m\}$ for the environments and

$$\Delta_m := \left\{ w \in \mathbb{R}_+^m : \sum_{e=1}^m w_e = 1 \right\}.$$

All vector norms in the mathematical statements are Euclidean. In Lean they are represented by the explicit coordinate sums $\text{sqNorm}(x) = \sum_i x_i^2$ and $\text{sqDist}(x, y) = \sum_i (x_i - y_i)^2$, rather than by the default norm on the raw function type $\text{Fin}(p) \rightarrow \mathbb{R}$.

2.1 NegDRO population objectives

For an environmental pair $(X^e, Y^e) \in \mathbb{R}^p \times \mathbb{R}$, define

$$R_e(b) := \mathbb{E}[(Y^e - b^\top X^e)^2]. \quad (1)$$

The original negative-weight set and its simplex reparameterization are

$$\mathcal{U}(\gamma) := \left\{ u \in \mathbb{R}^m : \sum_e u_e = 1, \min_e u_e \geq -\gamma \right\}, \quad u_e = (1 + \gamma m)w_e - \gamma,$$

where $\gamma \geq 0$ and $w \in \Delta_m$. After division by the positive constant $1 + \gamma m$, set

$$a_\gamma := \frac{\gamma}{1 + \gamma m}, \quad (2)$$

$$\mathcal{L}(b, w) := \sum_{e=1}^m (w_e - a_\gamma) R_e(b), \quad \Phi(b) := \max_{w \in \Delta_m} \mathcal{L}(b, w), \quad (3)$$

$$\mathcal{L}_\mu(b, w) := \mathcal{L}(b, w) - \mu \|w\|_2^2, \quad \Phi_\mu(b) := \max_{w \in \Delta_m} \mathcal{L}_\mu(b, w). \quad (4)$$

Here $\mu \geq 0$ is the dual penalty. The population objectives originate in the NegDRO formulation of Wang et al. [1]; the stochastic recursion studied below is the extension developed in this note.

2.2 Corrected realized additive-intervention model

The structural equations are

$$Y = \beta^{*\top} X + \varepsilon_Y, \quad X = B_{YX}Y + B_{XX}X + \varepsilon_X. \quad (5)$$

The additive-intervention decomposition writes the realized noises as

$$\varepsilon_Y = \eta_Y, \quad \varepsilon_X = \eta_X + \delta_e. \quad (6)$$

The final Lean entry point starts from the resulting realized environmental maps, rather than reconstructing them from acyclicity or a block-inverse theorem. Let

$$S := (I - B_{XX}) - B_{YX}\beta^{*\top}, \quad G^\top S = I.$$

Direct substitution in (5) gives

$$SX = B_{YX}\varepsilon_Y + \varepsilon_X. \quad (7)$$

Consequently the corrected realized maps used by the final theorem are

$$X^e = G^\top (B_{YX}\eta_Y + \eta_X + \delta_e), \quad (8)$$

$$Y^e = \beta^{*\top} X^e + \eta_Y. \quad (9)$$

The Lean argument named **GT** represents the matrix denoted here by $G^\top$. The inverse identity $G^\top S = I$ explains the structural elimination above, but it is not a field of the final public assumption

bundle. That interface starts from the corrected realized maps (8)–(9); it does not take B_{XX} or acyclicity as parameters.

The primitive model assumptions require integrability of the coordinate products needed for the second-moment calculation: η_Y^2 , $\eta_Y \eta_{X,i}$, $\eta_Y \delta_{e,i}$, $\eta_{X,i} \eta_{X,j}$, $\eta_{X,i} \delta_{e,j}$, $\delta_{e,i} \eta_{X,j}$, and $\delta_{e,i} \delta_{e,j}$. They also require the systematic/intervention cross moments to vanish, in particular

$$\mathbb{E}[\eta_Y \delta_e] = 0, \quad \mathbb{E}[\eta_X \delta_e^\top] = 0, \quad (10)$$

with the corresponding coordinate orientations. These are substantive model assumptions in the final theorem.

Define the systematic covariate noise $U := B_{YX} \eta_Y + \eta_X$ and the common linear coefficient

$$v_{\text{SEM}} := -G^\top \mathbb{E}[U \eta_Y]. \quad (11)$$

Under (10), $\mathbb{E}[X^e \eta_Y] = -v_{\text{SEM}}$ for every e . With

$$\sigma_Y^2 := \mathbb{E}[\eta_Y^2], \quad \Sigma_e := \mathbb{E}[X^e X^{e\top}], \quad (12)$$

the environmental risk therefore has the common expansion

$$R_e(b) = \sigma_Y^2 + 2v_{\text{SEM}}^\top (b - \beta^*) + (b - \beta^*)^\top \Sigma_e (b - \beta^*). \quad (13)$$

The Lean development proves this identity from the actual squared-loss integral and the realized moment relations; it does not identify two intended “KL-like” or “risk-like” expressions by name alone.

Remark 2.1 (Source sign convention). Equation (10) of arXiv:2412.11850v3 forces the positive term $+G^\top B_{YX} \varepsilon_Y$ in the reduced form (8). Appendix E.1 Eq. (97) instead prints the opposite sign in the lower-left inverse block. The negative cross terms later printed in the matrix H are also inconsistent with the displayed positive-block factorization. The formalization follows the algebra implied by Eq. (10) and verifies the corrected positive-sign identity. It does not claim to verify the inconsistent printed Eq. (97). This is a source discrepancy, not an inference about the authors’ intent.

2.3 Heterogeneity witness and feasible geometry

For $w \in \Delta_m$, define

$$A(w) := \sum_{e=1}^m \left(w_e - \frac{1}{m} \right) \Sigma_e. \quad (14)$$

Assumption 2.2 (Fixed spectral witness). There are $w^0 \in \Delta_m$ and $\lambda > 0$ such that every eigenvalue of the symmetric matrix $A(w^0)$ is at least λ . Equivalently,

$$A(w^0) \succeq \lambda I_p. \quad (15)$$

The fixed vector w^0 is used only in the proof; it is not supplied to the algorithm.

Assumption 2.3 (Closed convex bounded feasible set). Let $C \subseteq \mathbb{R}^p$ be nonempty, closed, and convex, with

$$b_{\text{Init}}, \beta^* \in C, \quad \|b\|_2^2 \leq B^2 \quad \text{for every } b \in C, \quad B \geq 0. \quad (16)$$

Write P_C for the Euclidean nearest-point projection onto C .

The formalization constructs P_C by transporting coordinates to a Euclidean Hilbert space. It proves existence, membership in C , uniqueness, the variational inequality, the strong Pythagorean (Fejér) inequality, nonexpansiveness, continuity, and measurability. Thus the projection used in the final recursion is not an abstract Fejér interface. The ball $C = \{b : \|b\|_2 \leq B\}$ is one concrete example. Its classical explicit formula is

$$P_C(z) = \begin{cases} z, & \|z\|_2 \leq B, \\ Bz/\|z\|_2, & \|z\|_2 > B. \end{cases} \quad (17)$$

The Lean development formalizes the general nearest-point construction used by the theorem; it does not separately package (17) as a ball-specific theorem.

2.4 Fresh conditional sampling law

The stochastic process is defined on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ with a pre-round filtration $(\mathcal{F}_{t-1})_{t \geq 1}$. At round t , the observed environment and data are (E_t, X_t, Y_t) . Intuitively, E_t is conditionally uniform and the observation is fresh from its selected environmental law. The precise final assumption is the following law-level identity.

Assumption 2.4 (Universal fresh conditional law). For each used round t , environment e , and real test function φ that is integrable under the fixed environment law ν_e ,

$$\mathbb{E}[\mathbf{1}_{\{E_t=e\}}\varphi(X_t, Y_t) \mid \mathcal{F}_{t-1}] = \frac{1}{m} \int \varphi(X^e(\xi), Y^e(\xi)) d\nu_e(\xi) \quad \text{a.e.} \quad (18)$$

The selected test on the left is integrable, each ν_e is a probability measure, and the observed coordinates have the required a.e. strong measurability. In the final additive wrapper, all ν_e are the common noise law ν and (X^e, Y^e) are the realized maps (8)–(9).

Taking $\varphi = 1$ gives conditional uniform environment selection. Taking XX , XY , and Y^2 monomials gives the selected population moments; fourth degree tests give the raw moment inequalities used below. These identities, together with predictability of the current state, derive conditional unbiasedness of the actual oracles. Conditional oracle unbiasedness is not a separate final assumption.

The filtration is increasing, each conditioning sigma-algebra is contained in the ambient sigma-algebra, and the round- t primal and dual oracles are measurable with respect to the post-observation sigma-algebra $\mathcal{F}_t$. In Lean, this oracle measurability is assumed for every natural-number index, whereas the universal fresh law is required only for the used rounds $t \leq T$. The current iterates are consequently $\mathcal{F}_{t-1}$ -measurable. These adaptedness and measurability conditions remain explicit assumptions. The formalization does not construct a product probability space, a natural filtration, or an infinite independent stream.

2.5 Concrete oracles, moments, and integrability

Assume the environmental coordinates X_i^e and outcomes Y^e are a.e. strongly measurable. Separately, assume the environmental fourth moments are measurable and integrable and satisfy, for nonnegative K_X, K_Y ,

$$\mathbb{E}\|X^e\|_2^4 \leq K_X, \quad \mathbb{E}|Y^e|^4 \leq K_Y \quad \text{for every } e. \quad (19)$$

These assumptions do not require independence between X^e and Y^e .

At the predictable pair (b_t, w_t) , the concrete coordinate oracles are

$$[\hat{g}_t^b]_i = 2m \left(w_{t,E_t} - \frac{\gamma}{1 + \gamma m} \right) (X_t^\top b_t - Y_t) X_{t,i}, \quad (20)$$

$$[\hat{g}_t^{w,\mu}]_j = m(Y_t - X_t^\top b_t)^2 \mathbf{1}_{\{j=E_t\}} - 2\mu w_{t,j}. \quad (21)$$

The unpenalized dual oracle is the specialization $\hat{g}_t^{w,0}$. The universal law gives, a.e.,

$$\mathbb{E}[\hat{g}_t^b \mid \mathcal{F}_{t-1}] = \nabla_b \mathcal{L}_\mu(b_t, w_t), \quad (22)$$

$$\mathbb{E}[\hat{g}_t^{w,\mu} \mid \mathcal{F}_{t-1}] = \nabla_w \mathcal{L}_\mu(b_t, w_t). \quad (23)$$

The first equality is independent of μ because the penalty contains no b .

Here the population-gradient notation abbreviates the explicit finite-coordinate vector used in the formalization. Lean proves its coordinate formula, the pairing identities required by the recursion, and the corresponding radial `HasDerivAt` statement; it does not expose a separate full Fréchet-gradient theorem. No stronger differentiability claim is needed for the convergence inequalities below.

The exact squared-moment constants used by the final theorems are

$$G_b^2 := 4m \left(2\sqrt{K_Y K_X} + 2B^2 K_X \right), \quad (24)$$

$$G_{w,0}^2 := m^2 (8K_Y + 8B^4 K_X), \quad (25)$$

$$G_{w,\mu}^2 := 2m^2 (8K_Y + 8B^4 K_X) + 8\mu^2. \quad (26)$$

These symbols already denote squared bounds and are never squared again. The derivation uses

$$\begin{aligned} (X^\top b - Y)^2 \|X\|_2^2 &\leq 2\|X\|_2^2 Y^2 + 2B^2 \|X\|_2^4, \\ (Y - X^\top b)^4 &\leq 8Y^4 + 8B^4 \|X\|_2^4, \end{aligned}$$

the simplex bounds on w_t , and the fresh selected moments. It yields the conditional estimates

$$\mathbb{E}[\|\hat{g}_t^b\|_2^2 \mid \mathcal{F}_{t-1}] \leq G_b^2, \quad (27)$$

$$\mathbb{E}[\|\hat{g}_t^{w,0}\|_\infty^2 \mid \mathcal{F}_{t-1}] \leq G_{w,0}^2, \quad (28)$$

$$\mathbb{E}[\|\hat{g}_t^{w,\mu}\|_\infty^2 \mid \mathcal{F}_{t-1}] \leq G_{w,\mu}^2. \quad (29)$$

Here the dual infinity bound is proved coordinatewise; the Lean development does not identify it definitionally with the default function-space norm.

Oracle integrability is derived, not assumed. Fresh selected mixed and fourth moments first give integrability of $\|\hat{g}_t^b\|_2^2$. Coordinate domination $|\hat{g}_t^b|_i|^2 \leq \|\hat{g}_t^b\|_2^2$ and the probability-space implication $L^2 \subseteq L^1$ give primal coordinate integrability. The dual coordinate-bound process is controlled by

$$m(Y_t - X_t^\top b_t)^2 + 2\mu;$$

the residual fourth-moment estimate gives integrability of its square, and the same $L^2 \subseteq L^1$ argument gives dual coordinate integrability. Trajectory feasibility, simplex invariance, and oracle measurability supply the adaptive parts of these arguments.

2.6 Coupled projected-SGD/EG recursion and indexing

We use the paper-style convention $t = 1, \dots, T$. Lean uses the zero-based range $r = 0, \dots, T - 1$; the note's (b_t, w_t) corresponds to Lean's state at $r = t - 1$. Initialize

$$b_1 = b_{\text{Init}} \in C, \quad w_1^{\text{alg}} = m^{-1} \mathbf{1}. \quad (30)$$

The superscript distinguishes the algorithmic uniform initialization from the fixed comparator w^0 . For constant positive steps η_b, η_w , define

$$b_{t+1} = P_C(b_t - \eta_b \hat{g}_t^b), \quad (31)$$

$$w_{t+1,j} = \frac{w_{t,j} \exp(\eta_w [\hat{g}_t^{w,\mu}]_j)}{\sum_k w_{t,k} \exp(\eta_w [\hat{g}_t^{w,\mu}]_k)}. \quad (32)$$

Thus the primal step is descent and the dual step is ascent. Both oracles use the same current state and may use the same round sample. The proof requires predictability of the displacement, not independence between the primal and dual stochastic gradients.

The two error summaries used below are

$$D_T := \frac{1}{T} \sum_{t=1}^T \mathbb{E} \|b_t - \beta^*\|_2^2, \quad \bar{b}_T := \frac{1}{T} \sum_{t=1}^T b_t. \quad (33)$$

The final assumption interface additionally requires $m, T > 0$, $\eta_b, \eta_w, \lambda > 0$, and $\gamma \geq 0$. Penalized results require $\mu \geq 0$.

3 Penalized NegDRO

For $\mu \geq 0$, recall $\mathcal{L}_\mu(b, w) = \mathcal{L}(b, w) - \mu \|w\|_2^2$. When $\mu > 0$, the inner population objective is strongly concave and has a unique maximizer. Official Algorithm 1 of Wang et al. [1] computes the corresponding maximizer of an empirical objective before its primal step. The algorithm in this note instead applies the simultaneous fresh-sample recursion (31)–(32). Its proof never requires the online weight w_t to be an exact inner maximizer.

3.1 Fixed-witness direction and comparator identity

Let $d = b - \beta^*$ and define

$$\varepsilon_{\text{id}} := \frac{\|v_{\text{SEM}}\|_2^2}{\lambda(1 + \gamma m)^2}. \quad (34)$$

The risk expansion (13) and the spectral witness imply the following direction inequality.

Lemma 3.1 (Fixed-witness direction). *For every $b \in \mathbb{R}^p$,*

$$(b - \beta^*)^\top \nabla_b \mathcal{L}_\mu(b, w^0) \geq \lambda \|b - \beta^*\|_2^2 - \varepsilon_{\text{id}}. \quad (35)$$

Proof. The penalty is independent of b . Decomposing the weighted second-moment matrix into the positive-semidefinite average second-moment matrix and $A(w^0)$ gives

$$\frac{1}{2} d^\top \nabla_b \mathcal{L}_\mu(b, w^0) \geq \lambda \|d\|_2^2 - \frac{\|v_{\text{SEM}}\|_2^2}{1 + \gamma m} \|d\|_2.$$

Young's inequality absorbs one copy of the quadratic curvature, giving (35). Notice that the identification term at this direction stage is $\|v_{\text{SEM}}\|_2^2 / [\lambda(1 + \gamma m)^2]$. $\square$

Define the comparator-minus-current quantity

$$\text{Reg}_{t,\mu}^0 := \mathcal{L}_\mu(b_t, w^0) - \mathcal{L}_\mu(b_t, w_t). \quad (36)$$

It need not be nonnegative at a particular round.

Lemma 3.2 (Penalized comparator reduction). *For every round t ,*

$$(b_t - \beta^*)^\top \nabla_b \mathcal{L}_\mu(b_t, w_t) \geq \lambda \|b_t - \beta^*\|_2^2 - \varepsilon_{\text{id}} - 2 \text{Reg}_{t,\mu}^0 - 2\mu. \quad (37)$$

Proof. The common scalar identity in Appendix A gives

$$\begin{aligned} & (b_t - \beta^*)^\top \nabla_b \mathcal{L}_\mu(b_t, w_t) \\ &= (b_t - \beta^*)^\top \nabla_b \mathcal{L}_\mu(b_t, w^0) - 2 \text{Reg}_{t,\mu}^0 - 2\mu (\|w^0\|_2^2 - \|w_t\|_2^2). \end{aligned}$$

For simplex vectors, $\|w^0\|_2^2 \leq 1$ and $\|w_t\|_2^2 \geq 0$, so the last difference is at most one. Combining this fact with Lemma 3.1 proves the claim. This is the sole source of the structural 2μ loss. $\square$

3.2 Entropy regret and projected recursion

The exact exponentiated-gradient update remains strictly positive and in the simplex. The formalized entropy argument proves the three-point identity, finite Pinsker inequality

$$\frac{1}{2} \|p - q\|_1^2 \leq D_\Phi(p, q),$$

coordinate Hölder, and the scalar square completion

$$-\frac{1}{2}a^2 + ba \leq \frac{1}{2}b^2.$$

It also proves terminal Bregman nonnegativity for a possibly boundary comparator and the initial-potential bound $D_\Phi(w^0, m^{-1}\mathbf{1}) \leq \log m$. Therefore no Pinsker, mirror-step, initial-potential, or terminal-potential premise remains in the final theorem. With the conditional dual moment estimate from (29), the resulting expected regret is

$$\sum_{t=1}^T \mathbb{E}[\text{Reg}_{t,\mu}^0] \leq \frac{\log m}{\eta_w} + \frac{\eta_w T G_{w,\mu}^2}{2}. \quad (38)$$

The actual nearest-point projection satisfies, because $\beta^* \in C$,

$$\|b_{t+1} - \beta^*\|_2^2 \leq \|b_t - \eta_b \hat{g}_t^b - \beta^*\|_2^2.$$

Predictability and the a.e. conditional mean identity turn the cross term into the population direction. The conditional primal second-moment bound then gives

$$\mathbb{E}\|b_{t+1} - \beta^*\|_2^2 \leq \mathbb{E}\|b_t - \beta^*\|_2^2 - 2\eta_b \mathbb{E}\left[(b_t - \beta^*)^\top \nabla_b \mathcal{L}_\mu(b_t, w_t)\right] + \eta_b^2 G_b^2.$$

Substitution of Lemma 3.2 and telescoping complete the finite-time proof.

3.3 Verified penalized conclusions

The assumptions in the following results are precisely those stated in Section 2: the corrected realized model and primitive moments, the universal fresh conditional law and adaptedness conditions, the bounded closed convex feasible set, the simplex spectral witness, positive horizon and step sizes, and the ambient probability-measure condition. In addition, $\mu \geq 0$.

Theorem 3.3 (Penalized finite-time expected squared error). *The recursion (31)–(32) satisfies*

$$\begin{aligned} D_T \leq & \frac{\|b_{\text{Init}} - \beta^*\|_2^2}{2\lambda\eta_b T} + \frac{\|v_{\text{SEM}}\|_2^2}{\lambda^2(1 + \gamma m)^2} + \frac{2\mu}{\lambda} + \frac{2\log m}{\lambda\eta_w T} \\ & + \frac{\eta_w G_{w,\mu}^2}{\lambda} + \frac{\eta_b G_b^2}{2\lambda}. \end{aligned} \quad (39)$$

Theorem 3.4 (Penalized averaged-iterate expected squared error). *For $\bar{b}_T$ from (33),*

$$\begin{aligned} \mathbb{E}\|\bar{b}_T - \beta^*\|_2^2 \leq & \frac{\|b_{\text{Init}} - \beta^*\|_2^2}{2\lambda\eta_b T} + \frac{\|v_{\text{SEM}}\|_2^2}{\lambda^2(1 + \gamma m)^2} + \frac{2\mu}{\lambda} + \frac{2\log m}{\lambda\eta_w T} \\ & + \frac{\eta_w G_{w,\mu}^2}{\lambda} + \frac{\eta_b G_b^2}{2\lambda}. \end{aligned} \quad (40)$$

The second theorem follows from the first trajectory calculation together with finite-dimensional Jensen. It concerns the random averaged iterate, not the terminal state b_{T+1} after the T updates.

For the rate specialization, define

$$B_{\text{id}} := \frac{\|v_{\text{SEM}}\|_2^2}{\lambda^2(1 + \gamma m)^2}, \quad (41)$$

$$C_\mu := \frac{\|b_{\text{Init}} - \beta^*\|_2^2}{2\lambda} + \frac{2\log m}{\lambda} + \frac{G_{w,\mu}^2}{\lambda} + \frac{G_b^2}{2\lambda}. \quad (42)$$

Corollary 3.5 (Penalized inverse-square-root rate). *If $\eta_b = \eta_w = 1/\sqrt{T}$, then*

$$D_T \leq B_{\text{id}} + \frac{2\mu}{\lambda} + \frac{C_\mu}{\sqrt{T}}. \quad (43)$$

The $2\mu/\lambda$ term appears exactly once; there is no $2\mu T$ term. The corollary is an expected squared-error statement. Taking a square root gives an RMS consequence under the usual nonnegativity conditions, not a theorem about $\mathbb{E}\|b_t - \beta^*\|_2$.

4 Unpenalized NegDRO

The unpenalized population objective $\Phi(b) = \max_{w \in \Delta_m} \mathcal{L}(b, w)$ may be nonsmooth because the inner objective is linear in w . The stochastic extension does not attempt to select a maximizing w at every round. It applies (31)–(32) with $\mu = 0$ and controls static regret against the same fixed witness w^0 .

4.1 Direction and comparator reduction

Let ε_{id} be as in (34).

Lemma 4.1 (Unpenalized fixed-witness direction). *For every $b \in \mathbb{R}^p$,*

$$(b - \beta^*)^\top \nabla_b \mathcal{L}(b, w^0) \geq \lambda \|b - \beta^*\|_2^2 - \varepsilon_{\text{id}}. \quad (44)$$

The proof is the same fixed-witness curvature and vector Young inequality used in Lemma 3.1. Define

$$\text{Reg}_t^0 := \mathcal{L}(b_t, w^0) - \mathcal{L}(b_t, w_t). \quad (45)$$

The sign convention is comparator minus current, although the instantaneous quantity may be negative.

Lemma 4.2 (Unpenalized comparator reduction). *For every round t ,*

$$(b_t - \beta^*)^\top \nabla_b \mathcal{L}(b_t, w_t) \geq \lambda \|b_t - \beta^*\|_2^2 - \varepsilon_{\text{id}} - 2 \text{Reg}_t^0. \quad (46)$$

Proof. The common term in the loss–gradient identity is independent of w , so subtracting the identity at w^0 and w_t gives

$$(b_t - \beta^*)^\top \nabla_b \mathcal{L}(b_t, w_t) = (b_t - \beta^*)^\top \nabla_b \mathcal{L}(b_t, w^0) - 2 \text{Reg}_t^0.$$

Lemma 4.1 completes the proof. $\square$

The entropy mirror theorem described in Section 3 no longer takes Pinsker or the mirror-step conclusion as hypotheses. Under its stated simplex, positivity, and step-size conditions, it yields

$$\sum_{t=1}^T \mathbb{E}[\text{Reg}_t^0] \leq \frac{\log m}{\eta_w} + \frac{\eta_w T G_{w,0}^2}{2}. \quad (47)$$

Substitution into the same expected projected recursion gives the final unpenalized bounds.

4.2 Verified unpenalized conclusions

Theorem 4.3 (Unpenalized finite-time expected squared error). *Under the assumptions of Section 2, with $\mu = 0$, the recursion satisfies*

$$D_T \leq \frac{\|b_{\text{Init}} - \beta^*\|_2^2}{2\lambda\eta_b T} + \frac{\|v_{\text{SEM}}\|_2^2}{\lambda^2(1 + \gamma m)^2} + \frac{2 \log m}{\lambda\eta_w T} + \frac{\eta_w G_{w,0}^2}{\lambda} + \frac{\eta_b G_b^2}{2\lambda}. \quad (48)$$

Theorem 4.4 (Unpenalized averaged-iterate expected squared error). *The averaged iterate satisfies*

$$\mathbb{E}\|\bar{b}_T - \beta^*\|_2^2 \leq \frac{\|b_{\text{Init}} - \beta^*\|_2^2}{2\lambda\eta_b T} + \frac{\|v_{\text{SEM}}\|_2^2}{\lambda^2(1 + \gamma m)^2} + \frac{2 \log m}{\lambda\eta_w T} + \frac{\eta_w G_{w,0}^2}{\lambda} + \frac{\eta_b G_b^2}{2\lambda}. \quad (49)$$

The identification contribution in the direction inequality is ε_{id} ; division by λ in the primal recursion produces the displayed final bias $\|v_{\text{SEM}}\|_2^2/[\lambda^2(1 + \gamma m)^2]$.

Define the exact inverse-square-root coefficient

$$C_0 := \frac{\|b_{\text{Init}} - \beta^*\|_2^2}{2\lambda} + \frac{2 \log m}{\lambda} + \frac{G_{w,0}^2}{\lambda} + \frac{G_b^2}{2\lambda}. \quad (50)$$

Corollary 4.5 (Unpenalized inverse-square-root rate). *If $\eta_b = \eta_w = 1/\sqrt{T}$, then*

$$D_T \leq B_{\text{id}} + \frac{C_0}{\sqrt{T}}. \quad (51)$$

This public rate theorem controls D_T , not the terminal state b_{T+1} . The separate averaged-iterate theorem retains the general positive step sizes and the same finite-time right-hand side; no additional averaged inverse-square-root public entry point is asserted here.

5 Discussion and Scope

5.1 Relation to the official NegDRO algorithms

The verified theorem and the official algorithms answer related but distinct questions. The official algorithms form empirical environmental risks from a fixed data set, solve an inner maximization, and select an iterate using a stationarity criterion. The extension in this note targets population risks through a fresh conditional law and updates both variables online.

Feature	Official empirical method	Verified stochastic extension
Data access	Fixed environment data sets, reused across iterations	One conditionally fresh environmental observation per round
Dual operation	Exact empirical-risk maximization in w	One exponentiated-gradient ascent step
Primal operation	Gradient or subgradient step for the empirical envelope	Euclidean nearest-point-projected stochastic descent
Output analyzed	An iterate selected by a stationarity criterion	D_T and the random averaged iterate $\bar{b}_T$
Guarantee	Official finite-sample and optimization results in Wang et al. [1]	Conditional expected squared-error inequalities in Theorems 3.3– 4.4

This comparison does not imply a uniform sample- or computation-complexity ordering. In the official method, an iteration may revisit a fixed data set. Here T counts both online updates and fresh-law rounds. The present Lean development is therefore not a literal formalization of official arXiv:2412.11850v3 Algorithm 1.

5.2 What the fixed witness accomplishes

The analytical weight w^0 separates identification from the algorithm’s dual optimization. Spectral curvature at w^0 gives the primal direction bound, while the comparator identity transfers the difference between w^0 and w_t into a dual objective regret. Exponentiated-gradient ascent controls the cumulative comparator term even though an individual regret can be negative. The projected primal recursion then converts this cumulative control into an expected squared-error bound.

The penalized and unpenalized calculations differ in two explicit places. The dual oracle and its moment constant change from $G_{w,0}^2$ to $G_{w,\mu}^2$, and the simplex norm comparison contributes the structural bias $2\mu/\lambda$. There is no accumulation into $2\mu T$ after averaging.

5.3 Scope boundaries

Target A of the formalization is complete: the conditional theorem for the stochastic extension is kernel checked relative to its stated hypotheses. The following are genuine scope boundaries, not missing steps in that proof.

1. *Sampling realization.* The universal fresh conditional law (18) is assumed. The project does not construct a product probability space, a natural filtration, or an infinite independent stream.
2. *Reduced-form entry point.* The final theorem begins with the corrected realized maps (8)–(9). Separate Lean lemmas justify structural elimination from (5) when the required inverse identity is supplied, but the final wrapper does not derive $G^\top$ from B_{XX} and acyclicity.
3. *Expectation.* The results are in expectation. No high-probability theorem or adaptive concentration result is claimed.

4. *Output.* The analysis controls the time average D_T and the expected squared error of $\bar{b}_T$. It does not establish a last-iterate guarantee.
5. *Identification bias.* The term B_{id} remains unless v_{SEM} vanishes or is controlled through other structural information. The theorem does not silently identify it with an optimization error.
6. *Algorithmic scope.* This fresh-sample simultaneous projected SGD/EG method is not official v3 Algorithm 1 and does not verify exact empirical-risk maximization.
7. *Rate interface.* The project proves explicit inequalities. It does not add a formal `IsBigO` wrapper, and the RMS square-root consequence is not an expected-norm theorem.

The fourth-moment, bounded-domain, spectral heterogeneity, sign, and fresh-law conditions remain substantive theorem hypotheses. Calling the conditional proof complete does not mean that the theorem is assumption-free.

5.4 Conclusion

The corrected additive model, fixed spectral witness, conditional fresh-law oracle, entropy geometry, and Euclidean projection fit into one closed proof chain. The resulting finite-time inequalities separate three effects: initialization and online optimization, the structural identification term B_{id} , and—only in the penalized case— $2\mu/\lambda$. This yields a self-contained mathematical account of the finite-dimensional stochastic extension and a precise boundary between that extension and the original empirical NegDRO algorithms.

6 Lean Verification Status

The formalization is implemented in Lean 4 [3] on top of Mathlib [4]. Target A—the complete conditional convergence theorem for the finite-dimensional corrected-additive-model, fresh-sample, nearest-point-projected SGD/EG extension—is complete.

6.1 Checked public interface

The root imports all 50 project modules. The six publication-level entry points are defined in `AdditiveInterventionSamplingProjectedConvergence.lean`:

1. `stochasticNegDRO_additiveSampling_projected_unpenalized_convergence`;
2. `stochasticNegDRO_additiveSampling_projected_unpenalized_averaged_convergence`;
3. `stochasticNegDRO_additiveSampling_projected_unpenalized_invSqrt_rate`;
4. `stochasticNegDRO_additiveSampling_projected_penalized_convergence`;
5. `stochasticNegDRO_additiveSampling_projected_penalized_averaged_convergence`;
6. `stochasticNegDRO_additiveSampling_projected_penalized_invSqrt_rate`.

The first, second, fourth, and fifth correspond to the four theorem statements in Sections 3 and 4; the other two are the displayed inverse-square-root corollaries. The repository’s detailed mathematical dependency map is `docs/THEOREM_MAP.md`.

The final projected assumption structure has 30 fields: eight model and reduced-form fields, five sampling and adaptedness fields, and seventeen geometry and optimization fields. It also uses the ambient `IsProbabilityMeasure` instance, and penalized conclusions require $\mu \geq 0$. The model fields encode primitive product integrability, systematic/intervention orthogonality, environmental measurability, and fourth-moment integrability and bounds. The sampling fields encode filtration monotonicity, ambient inclusion, the universal fresh law, and next-round measurability of the two concrete oracles. The remaining fields encode the closed convex bounded feasible geometry, initialization and target feasibility, $m, T > 0$, positive steps, parameter signs, the simplex witness, and its genuine eigenvalue lower bound.

Coordinatewise primal and dual integrability, primal squared-norm integrability, and integrability of the squared dual coordinate-bound process are not fields of this public interface. They are derived in `OracleIntegrabilityBridge.lean` from selected fourth moments, bounded trajectory feasibility, simplex invariance, and measurability.

6.2 Local build and trust boundary

The checked working tree pins

- Lean v4.34.0-rc2; and
- Mathlib commit 85e3a25e006c35636f0e53b0e9296caca2685bc0.

Running `lake build` locally completes successfully with 8814 jobs. The root imports 50 out of 50 project source modules. A full Lean-source scan finds none of `sorry`, `admit`, `sorryAx`, custom axiom declarations, `unsafe`, `opaque`, `native_decide`, `implemented_by`, or `extern`.

Local `#print axioms` checks on the central oracle-integrability, Euclidean-projection, and six final convergence/rate theorems report only `propext`, `Classical.choice`, and `Quot.sound`. These are the standard foundations reached through Mathlib in this development. The results above describe a checked local working tree; they are not a claim of independent third-party reproduction or peer review. The model, sampling, moment, spectral, and feasible-set conditions remain mathematical hypotheses of the theorem.

AI tools assisted with library-API discovery, translation of paper algebra into Lean statements, refactoring, debugging, and adversarial source-semantic checks. AI-generated text or code was not treated as verification. Claims were retained only after elaboration, kernel checking, full builds, source scans, and repeated comparison of the public declarations with their mathematical interpretation.

A Proof Details

This appendix records the algebraic interfaces used by both final theorems. Each component described here is instantiated in the Lean proof chain; none is left as a premise of the six public projected entry points.

A.1 Common population identities

Write $d := b - \beta^*$ and

$$\bar{\Sigma} := \frac{1}{m} \sum_{e=1}^m \Sigma_e, \quad Q(w) := A(w) + \frac{\bar{\Sigma}}{1 + \gamma m}.$$

Because $\sum_e(w_e - a_\gamma) = 1/(1 + \gamma m)$, the corrected risk expansion gives

$$\mathcal{L}(b, w) = \frac{\sigma_Y^2 + 2v_{\text{SEM}}^\top d}{1 + \gamma m} + d^\top Q(w)d, \quad (52)$$

$$\nabla_b \mathcal{L}(b, w) = \frac{2v_{\text{SEM}}}{1 + \gamma m} + 2Q(w)d. \quad (53)$$

Hence

$$\mathcal{L}(b, w) = \frac{1}{2}d^\top \nabla_b \mathcal{L}(b, w) + \frac{\sigma_Y^2 + v_{\text{SEM}}^\top d}{1 + \gamma m}. \quad (54)$$

Since the penalty is independent of b ,

$$\mathcal{L}_\mu(b, w) = \frac{1}{2}d^\top \nabla_b \mathcal{L}_\mu(b, w) + \frac{\sigma_Y^2 + v_{\text{SEM}}^\top d}{1 + \gamma m} - \mu \|w\|_2^2. \quad (55)$$

These are the scalar identities formalized for the comparator reduction in this note. Their role is that the common term is identical at w^0 and w_t and therefore cancels in the subtraction. They should not be read as a renumbering of Eqs. (64)–(67) in the official appendix, whose displayed statements serve different purposes.

A.2 Curvature and comparator signs

At w^0 , $A(w^0) \succeq \lambda I$ and $\bar{\Sigma} \succeq 0$. Consequently

$$\begin{aligned} d^\top \nabla_b \mathcal{L}(b, w^0) &\geq 2\lambda \|d\|_2^2 - \frac{2\|v_{\text{SEM}}\|_2}{1 + \gamma m} \|d\|_2 \\ &\geq \lambda \|d\|_2^2 - \frac{\|v_{\text{SEM}}\|_2^2}{\lambda(1 + \gamma m)^2}. \end{aligned}$$

The Lean proof derives this from the actual symmetric second-moment matrices, the Hermitian eigenvalue lower bound, and a finite-coordinate Young inequality.

For the unpenalized comparator, subtraction of (54) gives

$$d_t^\top \nabla_b \mathcal{L}(b_t, w_t) = d_t^\top \nabla_b \mathcal{L}(b_t, w^0) - 2\text{Reg}_t^0. \quad (56)$$

For the penalized comparator, (55) gives

$$\begin{aligned} d_t^\top \nabla_b \mathcal{L}_\mu(b_t, w_t) &= d_t^\top \nabla_b \mathcal{L}_\mu(b_t, w^0) - 2\text{Reg}_{t,\mu}^0 \\ &\quad - 2\mu(\|w^0\|_2^2 - \|w_t\|_2^2). \end{aligned} \quad (57)$$

The regret sign is thus comparator minus current. Only a cumulative upper bound is used.

A.3 Formalized entropy mirror-ascent chain

Let $\phi(x) = x \log x$ with the standard zero convention, and let D_Φ be the coordinate Bregman divergence generated by $\sum_i \phi(w_i)$. For positive simplex vectors it equals the finite natural- logarithm relative entropy. The formalized finite Pinsker theorem has the normalization

$$\frac{1}{2} \left(\sum_i |p_i - q_i| \right)^2 \leq D_\Phi(p, q). \quad (58)$$

There is no factor $1/2$ in the definition of the ℓ_1 distance.

The normalized exponentiated-gradient ascent update is strictly positive and preserves the simplex. For a coordinate bound $|g_i| \leq G$, finite Hölder gives

$$|\langle g, x \rangle| \leq G \sum_i |x_i|.$$

This is the finite-coordinate specialization of the standard entropy mirror-ascent analysis; see, for example, Shalev-Shwartz [2]. The project proves the specialization directly rather than importing its conclusion as an assumption. Combining the exact three-point identity, (58), and

$$-\frac{1}{2}a^2 + ba \leq \frac{1}{2}b^2$$

proves the one-step inequality

$$\eta_w \langle g, u - w \rangle \leq D_\Phi(u, w) - D_\Phi(u, w^+) + \frac{\eta_w^2 G^2}{2}. \quad (59)$$

The comparator u need only belong to the simplex; it need not be strictly positive. A separate tangent inequality proves $D_\Phi(u, w^+) \geq 0$ for such a boundary comparator because w^+ is positive.

Summing (59) for the actual trajectory, using the uniform initialization, and applying the initial-potential bound

$$D_\Phi(u, m^{-1}\mathbf{1}) \leq \log m$$

yields the pathwise estimate

$$\sum_{t=1}^T \langle g_t, u - w_t \rangle \leq \frac{\log m}{\eta_w} + \frac{\eta_w}{2} \sum_{t=1}^T G_t^2. \quad (60)$$

The predictable conditional-expectation bridge proves, coordinatewise and then after finite summation, that

$$\mathbb{E}[\langle \hat{g}_t, d_t \rangle \mid \mathcal{F}_{t-1}] = \langle F_t, d_t \rangle \quad \text{a.e.},$$

and hence that the corresponding unconditional integrals are equal. It does not identify the two raw pairings a.e. This is the formal route to (38) and (47); no independence between the two oracle vectors enters.

A.4 Projection and regret-to-error reduction

For a nonempty closed convex C , the constructed nearest-point map satisfies the strong inequality

$$\|P_C(x) - z\|_2^2 + \|x - P_C(x)\|_2^2 \leq \|x - z\|_2^2, \quad z \in C. \quad (61)$$

Dropping the nonnegative middle term gives the Fejér inequality used in the recursion. Measurability of P_C and the post-observation oracles proves trajectory predictability by induction.

The common scalar reduction can be stated as follows.

Proposition A.1 (Comparator regret implies averaged squared error). *Suppose*

$$\begin{aligned} \mathbb{E}\|b_{t+1} - \beta^*\|_2^2 &\leq \mathbb{E}\|b_t - \beta^*\|_2^2 - 2\eta_b \mathbb{E}[(b_t - \beta^*)^\top F_t] + \eta_b^2 G_b^2, \\ (b_t - \beta^*)^\top F_t &\geq \lambda \|b_t - \beta^*\|_2^2 - \varepsilon - \delta - 2r_t, \end{aligned}$$

and $\sum_{t=1}^T \mathbb{E}[r_t] \leq \mathcal{R}_T$. Then

$$D_T \leq \frac{\|b_{\text{Init}} - \beta^*\|_2^2}{2\lambda\eta_b T} + \frac{\varepsilon + \delta}{\lambda} + \frac{2\mathcal{R}_T}{\lambda T} + \frac{\eta_b G_b^2}{2\lambda}. \quad (62)$$

The same right-hand side bounds $\mathbb{E}\|\bar{b}_T - \beta^*\|_2^2$.

Proof. Insert the direction inequality into the one-step recursion, sum from 1 to T , discard the nonnegative terminal squared distance, and divide by $2\lambda\eta_b T$. Finite-dimensional Jensen gives the averaged-iterate statement. $\square$

For the unpenalized result take $\varepsilon = \varepsilon_{\text{id}}$, $\delta = 0$, $r_t = \text{Reg}_t^0$, and use (47). For the penalized result take $\delta = 2\mu$, $r_t = \text{Reg}_{t,\mu}^0$, and use (38). These substitutions give the exact final constants, including the single $2\mu/\lambda$ penalty term.

A.5 Rate normalization and proof boundary

For positive integer T ,

$$\frac{1}{(1/\sqrt{T})T} = \frac{1}{\sqrt{T}}.$$

Substitution of $\eta_b = \eta_w = 1/\sqrt{T}$ in the two finite-time bounds therefore produces exactly $C_0/\sqrt{T}$ and $C_\mu/\sqrt{T}$. The Lean source proves this normalization as a real-number identity rather than invoking asymptotic notation.

The proof audit covers the corrected population expansion, comparator signs, finite Pinsker normalization, exponentiated-gradient regret, actual Euclidean projection, conditional predictable pairings, oracle moment and integrability bridges, stochastic recursion, spectral curvature, telescoping, Jensen, and rate algebra. It does not assert high-probability concentration, last-iterate convergence, construction of a canonical fresh sample stream, or verification of the official empirical exact-maximization algorithm.

B Source and Verification Map

This note uses arXiv:2412.11850v3, revised January 7, 2026, as the canonical source for the original NegDRO formulation and its empirical algorithms. The fresh-sample projected-SGD/EG theorem is an extension developed in this note. The table separates inherited source material, verified extension results, retained model assumptions, and out-of-scope constructions.

Component	Status	Source or checked Lean module
NegDRO objectives and empirical exact-maximization algorithms	Original-paper result	Wang et al. [1], especially official Algorithms 1–2
Positive-sign realized additive reduced form	Verified; source discrepancy	<code>AdditiveIntervention</code> <code>PopulationBridge.lean</code> ; follows Eq. (10), not the inconsistent printed Appendix E.1 Eq. (97)
Population risk and moment bridges	Verified	<code>PopulationRiskMomentBridge.lean</code> and <code>AdditiveIntervention</code> <code>SamplingBridge.lean</code>
Spectral witness to coordinate curvature	Verified under the stated eigenvalue hypothesis	<code>SpectralCurvatureBridge.lean</code> and <code>SpectralWitnessInterface.lean</code>
Universal fresh conditional law	Assumed model condition	<code>FreshSamplingLaw.lean</code> ; its moment and conditional-mean consequences are verified
Concrete oracle moments and integrability	Verified	<code>ConcreteOracle</code> <code>MomentBounds.lean</code> and <code>OracleIntegrabilityBridge.lean</code>
Entropy geometry, finite Pinsker, and EG regret	Verified	<code>FinitePinsker.lean</code> , <code>EGTrajectoryRegret.lean</code> , and <code>StochasticDualRegret.lean</code>
Euclidean nearest-point projection	Verified	<code>EuclideanProjectionBridge.lean</code>
Expected projected primal recursion and averaging	Verified	<code>StochasticPrimalTrajectory.lean</code> and <code>AveragedPrimalIterate.lean</code>
Six final projected convergence/rate entry points	Verified	<code>AdditiveIntervention</code> <code>SamplingProjected</code> <code>Convergence.lean</code>
Infinite i.i.d. stream or product-space construction	Out of scope	The universal fresh conditional law remains an explicit hypothesis
Literal verification of official v3 Algorithm 1	Out of scope	The verified recursion uses simultaneous fresh-sample projected SGD/EG
Formal <code>Asymptotics.IsBigO</code> wrapper	Out of scope	Exact finite inequalities and inverse-square-root identities are proved

The pinned implementation uses Lean `v4.34.0-rc2` and the following Mathlib commit:

`85e3a25e006c35636f0e53b0e9296caca2685bc0`.

The statements in this appendix describe the local checked working tree accompanying Version 1.0 of the note; they do not constitute independent third-party reproduction.

References

- [1] Zhenyu Wang, Yifan Hu, Peter Bühlmann, and Zijian Guo. *Causal Invariance Learning via Efficient Nonconvex Optimization*. arXiv:2412.11850v3, 2026. <https://arxiv.org/abs/2412.11850v3>.
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